

IA Generativa Multimodal

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Laboratorio 1: Toonification

StyleGAN Network Blending [3]

With a technique called **layer swapping**, it is possible to adapt a StyleGAN network [2] to another domain.



Collect data from the domain

First, you need to collect a specific dataset of faces. This was scrapped on the web.



Train the model

- A StyleGAN model is then fine-tuned on the dataset.
- The more the model is fine-tuned, the more the faces looks like ukiyoe drawings.
- It needs to be stopped when it is between the domain.
- The 2nd image looks the best



Latent Space

Let's look at the images generated when moving the latent vector:



The model basically learns not to have frontal faces.

Model interpolation: averaging

You can just average the weights of the models (the initial and the domain fine-tuned one), but it does not work very well:



Layer swapping

Model merging

Taking different resolution layers from the different models and combining them Helps to avoid changing the pose (which is controlled by early, low resolution layers) by early, low resolution layers)



Swapping in only the lower resolution layers from FFHQ into Ukiyo-e serves to preserve the pose of the generated face, but still transfer the features and style of a typical ukiyo-e portrait. Here's a detail of the point at which I think it looks best.

Toonify yourself!



Figure 1: Database of cartoon faces

Toonify yourself!



Figure 1: Fine-tuned StyleGAN model

Toonify yourself!

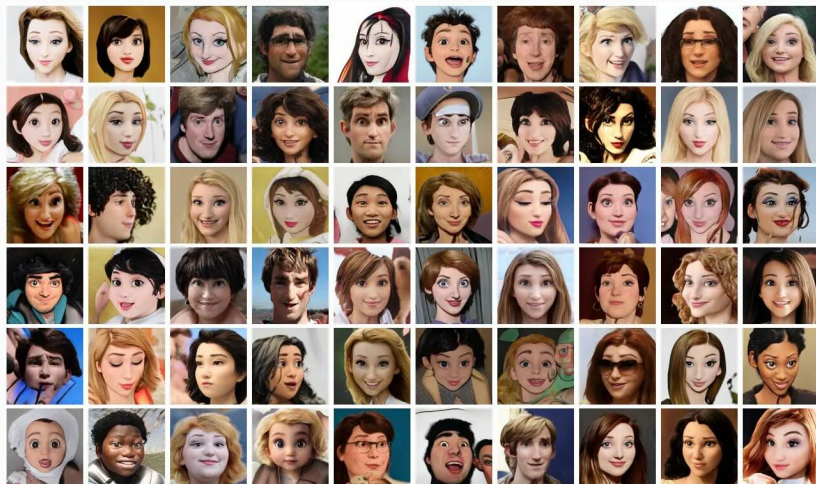


Figure 1: Results after Layer Swapping

Toonify yourself! Gather data, learn new model, merge it.

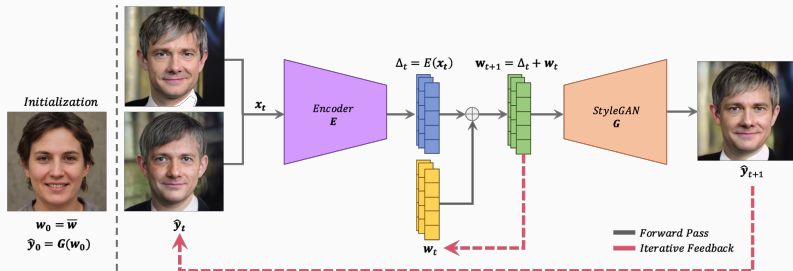


Toonify yourself! Gather data, learn new model, merge it.



Images and models from [Justin Pinkey's website](#).

Using Restyle trick for image inversion [1]



Finding the Stylegan's latent space vector that generates a specific image can be quite long. Restyle helps to converge to it in a few iterations.

Questions?



Y. Alaluf, O. Patashnik, and D. Cohen-Or.

ReStyle: A Residual-Based StyleGAN Encoder via Iterative Refinement.

8 2021.



T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila.

Analyzing and Improving the Image Quality of StyleGAN.

3 2020.



J. Pinkney and D. Adler.

Resolution Dependent GAN Interpolation for Controllable Image Synthesis Between Domains.

In 4th Workshop on Machine Learning for Creativity and Design at NeurIPS 2020, volume 27, pages 8–18, 11 2020.